

Direct Proofs

Sehyun Kwon

Mathematical Data Science
Hanyang University ERICA

Review

Let U be a universe and let $P(x), Q(x)$ be predicates on U . A theorem often has the form

$$(\forall x \in U)(P(x) \Rightarrow Q(x)).$$

Read this as: “For every x in U , if $P(x)$ is true, then $Q(x)$ is true.”

Our goal is to prove this whole statement. We must rule out any element for which $P(x)$ is true but $Q(x)$ is false.

A proof uses definitions, established results, and valid inference steps.

This lecture develops direct proof and proof by contraposition.

Direct proof

To prove $P \Rightarrow Q$ directly, we do not try to prove that P is actually true. Instead, we reason under the assumption that P is true:

Assume P .

From this assumption,

1. use definitions and established results;
2. apply valid inference steps;
3. derive Q .

If this reasoning derives Q , we conclude

$$\begin{array}{ccc} P & \text{temporary assumption} & \\ \vdots & & \\ Q & \implies & P \Rightarrow Q. \end{array}$$

Our final conclusion is $P \Rightarrow Q$. We have shown Q under the assumption P , not Q without that assumption.

Why direct proof works

The implication $P \Rightarrow Q$ is false in exactly one truth assignment:

$$P = \mathbf{T}, \quad Q = \mathbf{F}.$$

In a direct proof, we show that whenever P is true, Q must also be true. One example with both true would not be enough.

Therefore the only assignment that could make the implication false is impossible:

$$\underbrace{P = \mathbf{T}}_{\text{assumed}} \quad \Longrightarrow \quad \underbrace{Q = \mathbf{T}}_{\text{derived}}.$$

If $P = \mathbf{F}$, then $P \Rightarrow Q$ is already true by the definition of implication.

A direct proof rules out the only way $P \Rightarrow Q$ could be false.

Example 1: An even square

Theorem. For every integer x , if x is even, then x^2 is even.

Definitions. An integer is even if it is $2k$ for some integer k . For $x \in \mathbb{Z}$, define

$$P(x) = "x \text{ is even}", \quad Q(x) = "x^2 \text{ is even"}.$$

Quantified statement

$$(\forall x \in \mathbb{Z})(P(x) \Rightarrow Q(x)).$$

Proof plan. Assume $P(x)$ and use the definition of evenness to establish $Q(x)$.

Example 1: Proof

Claim: For every integer x , if x is even, then x^2 is even.

Proof.

Assumption. Let $x \in \mathbb{Z}$ be arbitrary and assume $P(x)$: x is even.
By definition, $x = 2k$ for some integer k .

Goal. Prove $Q(x)$ by writing x^2 as twice an integer:

$$x^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Since $2k^2$ is an integer, the last expression has the form $2m$ for an integer m .

Conclusion. $Q(x)$ holds. Since x was arbitrary, the quantified statement is proved. □

Example 2: A square-root inequality

Theorem. For positive real numbers n, a , if $a > \sqrt{n}$, then $n/a < \sqrt{n}$.

Definitions. Let $U = \{t \in \mathbb{R} : t > 0\}$. For $n, a \in U$, define

$$P(n, a) = "a > \sqrt{n}", \quad Q(n, a) = "n/a < \sqrt{n}".$$

Quantified statement

$$(\forall n \in U)(\forall a \in U)(P(n, a) \Rightarrow Q(n, a)).$$

Proof plan. Assume $P(n, a)$ and divide by positive quantities to obtain $Q(n, a)$.

Example 2: Proof

Claim: For positive n, a , $a > \sqrt{n}$ implies $n/a < \sqrt{n}$.

Proof.

Assumption. Let $n, a \in U$ be arbitrary and assume $P(n, a)$: $a > \sqrt{n}$.
Thus $a > \sqrt{n} > 0$.

Goal. Establish $Q(n, a)$: $n/a < \sqrt{n}$.

Since $\sqrt{n} > 0$, divide $a > \sqrt{n}$ by the positive number $a\sqrt{n}$:

$$\frac{1}{\sqrt{n}} > \frac{1}{a}.$$

Multiplying by the positive number n gives

$$\frac{n}{a} < \frac{n}{\sqrt{n}} = \sqrt{n}.$$

Conclusion. $Q(n, a)$ holds for every allowed pair satisfying $P(n, a)$, proving the quantified statement. □

Example 3: Divisibility by three

Theorem. If a positive integer's digit sum is divisible by 3, then the integer is divisible by 3.

Definitions. Let $U = \{1, 2, 3, \dots\}$, and let $s(n)$ be the sum of the decimal digits of n . For example, $s(372) = 3 + 7 + 2 = 12$. The notation $3 \mid t$ means $t = 3q$ for some integer q . Define on U :

$$P(n) = "3 \mid s(n)", \quad Q(n) = "3 \mid n".$$

Quantified statement

$$(\forall n \in U)(P(n) \Rightarrow Q(n)).$$

Proof plan. Show that $n - s(n)$ is a multiple of 3.

Example 3: Decimal notation

Our goal is to relate a positive integer n to its digit sum $s(n)$.

For example, 372 has digit sum $3 + 7 + 2 = 12$, and $372 = 3 \cdot 124$. We will prove the claim for every positive integer, not just this example.

Let n have $r + 1$ digits, with $r \geq 0$. Write them as $a_0, \dots, a_r$, starting from the units digit. For 372, $a_0 = 2$, $a_1 = 7$, and $a_2 = 3$.

Then

$$n = a_0 + a_1 10 + a_2 10^2 + \cdots + a_r 10^r,$$

where each $a_j \in \{0, 1, \dots, 9\}$.

The **digit sum** of n is

$$s(n) = a_0 + a_1 + \cdots + a_r.$$

Each digit multiplies a power of 10. We next express each power of 10 as 1 plus a multiple of 3.

Example 3: A supporting lemma

We need to connect the digit sum with the powers of 10 in n . A **lemma** is a supporting result used in a proof.

Lemma

For every integer $j \geq 0$, the integer $10^j - 1$ is divisible by 3.

For $j = 0$, $10^0 - 1 = 0$, which is divisible by 3.

For $j \geq 1$,

$$10^j - 1 = 9(1 + 10 + 10^2 + \cdots + 10^{j-1}).$$

Multiplying the sum by $10 - 1$ cancels all the middle terms, leaving $10^j - 1$.

The right-hand side is 3 times an integer, so $3 \mid (10^j - 1)$.

Thus, for each j , there is an integer m_j such that

$$10^j = 1 + 3m_j.$$

Example 3: Proof

Claim: For every positive integer n , $3 \mid s(n)$ implies $3 \mid n$.

Assumption. Let $n \in U$ be arbitrary and assume $P(n)$. Thus $s(n) = 3q$ for some integer q .

Goal. Prove $Q(n)$ by writing n as 3 times an integer.

Substitute into the decimal expansion and collect the digit terms:

$$\begin{aligned} n &= a_0 + a_1(1 + 3m_1) + \cdots + a_r(1 + 3m_r) \\ &= s(n) + 3(a_1m_1 + \cdots + a_rm_r). \end{aligned}$$

Thus n differs from its digit sum by a multiple of 3.

Hence

$$n = 3(q + a_1m_1 + \cdots + a_rm_r).$$

Conclusion. The parentheses contain an integer, so $Q(n)$ holds. Since n was arbitrary, the quantified statement is proved.

Proof by contraposition

Goal: Prove $P \Rightarrow Q$. The implication $P \Rightarrow Q$ is logically equivalent to its contrapositive:

$$(P \Rightarrow Q) \equiv (\neg Q \Rightarrow \neg P).$$

Therefore, to prove $P \Rightarrow Q$, we may instead:

1. assume that Q is false, and
2. derive that P is false.

This method is called **proof by contraposition**.

Both implications fail in exactly the same case: P true and Q false. Proving the contrapositive therefore proves the original claim.

Odd integers

An integer x is **odd** if

$$x = 2k + 1$$

for some integer k .

Every integer is either even or odd, but not both.

We will prove:

Theorem

For every integer x , if x^2 is even, then x is even.

Its contrapositive is: if x is odd, then x^2 is odd.

Over the integers, “not even” means “odd.” Assuming x is odd gives a useful expression for x to square.

Even square implies even integer

Original claim: If an integer x has an even square, then x is even.

Proof.

We prove: if x is odd, then x^2 is odd. Let x be an arbitrary integer and assume that x is odd.

Then $x = 2k + 1$ for some integer k .

Our goal is to express x^2 as $2m + 1$ for an integer m :

$$\begin{aligned}x^2 &= (2k + 1)^2 \\&= 4k^2 + 4k + 1 \\&= 2(2k^2 + 2k) + 1.\end{aligned}$$

Since $2k^2 + 2k$ is an integer, x^2 is odd.

Thus an odd integer cannot have an even square. By contraposition, if x^2 is even, then x is even. □

Proving a biconditional

To prove $P \Leftrightarrow Q$, prove both implications:

$$P \Rightarrow Q \quad \text{and} \quad Q \Rightarrow P.$$

A proof of one direction alone does not prove the reverse direction.
We have proved

$$x \text{ even} \Rightarrow x^2 \text{ even}$$

directly, and

$$x^2 \text{ even} \Rightarrow x \text{ even}$$

by contraposition.

Therefore, for every integer x ,

$$x \text{ is even} \Leftrightarrow x^2 \text{ is even.}$$

Assuming the desired conclusion

Consider the false claim: “For all real numbers x and y , $x = y$.”

The following argument begins by assuming $x = y$ and derives $0 = 0$:

$$x = y \Rightarrow 0x = 0y \Rightarrow 0 = 0.$$

The argument only proves the true implication

$$x = y \Rightarrow 0 = 0.$$

It does not derive $x = y$ for arbitrary x and y . Assuming the desired conclusion cannot establish it.

For example, $x = 1$, $y = 2$ makes $0 = 0$ true but $x = y$ false. Reaching a true statement does not justify reversing the reasoning.

Reading and writing proofs

When reading a proof, check three levels:

1. Does each step follow from definitions, earlier steps, or known results?
2. Why is each step useful for reaching the conclusion?
3. Can the same strategy be applied to another theorem?

A proof should also state the universe, assumptions, and conclusion clearly.

Proof writing improves through reading correct proofs and constructing new ones.

Summary

- ▶ A direct proof assumes the hypothesis and derives the conclusion.
- ▶ A universal proof begins with an arbitrary element of the universe.
- ▶ A proof by contraposition proves $\neg Q \Rightarrow \neg P$ instead of $P \Rightarrow Q$.
- ▶ A biconditional requires proofs in both directions.
- ▶ Every proof step must be justified and must support the stated conclusion.